

Continuous Assessment Test (CAT) – II – September 2026

Programme	: B. Tech (VLSI)	Semester	: Fall 25-26
Course Code & Course Title	: BEVD301L & ASIC DESIGN	Class Number	: CH2026270101934
Faculty	: Dr. Sakthivel.SM	Slot	: A1+TA1
Duration	: 90 Minutes	Max. Mark	: 50 Marks

General Instructions: Write only your registration number on the question paper in the box provided and do not write other information

Answer all questions

Part A (5 x 2 = 10 Marks)

Q. No Sub Sec.	Description	Marks	CO	BT Lev
1	Explain the concept of a multicycle path in Static Timing Analysis (STA). Why is an $N - 1$ hold correction required when an N -cycle multicycle constraint is applied? Illustrate the setup and hold edge relationships with a suitable example.	5	CO3	L4
2	Explain the operation of an asynchronous FIFO for reliable transfer of multi-bit data between two unrelated clock domains. Discuss the roles of the write and read pointers, pointer synchronization, and the generation of FIFO full and empty conditions. Why is Gray-code representation preferred for transferring FIFO pointers across clock domains? [5 Marks]	5	CO4	L2

Part B (1 x 10 = 10 Marks)

3	Consider the following RTL description of a synchronous datapath: <pre> module design (input wire clk, input wire rst_n, input wire [7:0] a, input wire [7:0] b, input wire [7:0] c, output reg [16:0] y); always @(posedge clk or negedge rst_n) begin if (!rst_n) y <= 17'd0; else y <= (a * b) + c; end endmodule </pre> For the above implementation, assume: $T_{MULT} = 5\text{ ns}$, $T_{ADD} = 2\text{ ns}$. Analyze the datapath and perform the following:	10	CO2	L4
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- Identify the critical combinational path and determine its delay. [2 Marks]
- Restructure the RTL by introducing an appropriate register to divide the critical combinational path into shorter stages while preserving the required computation. [2 Marks]
- Illustrate the datapath before and after restructuring, clearly indicating the register boundaries and combinational stages. [2 Marks]
- Neglecting register setup time, clock-to-Q delay, clock skew and clock uncertainty, calculate the maximum operating frequency (F_{max}) before and after restructuring. [2 Marks]
- Discuss the effect of the modified architecture on latency, area and power consumption. [2 Marks]

Part c (2 x 15 = 30 Marks)

- 4 Consider the synchronous circuit shown below in Fig.1. All flip-flops are positive-edge triggered and driven by the same clock.

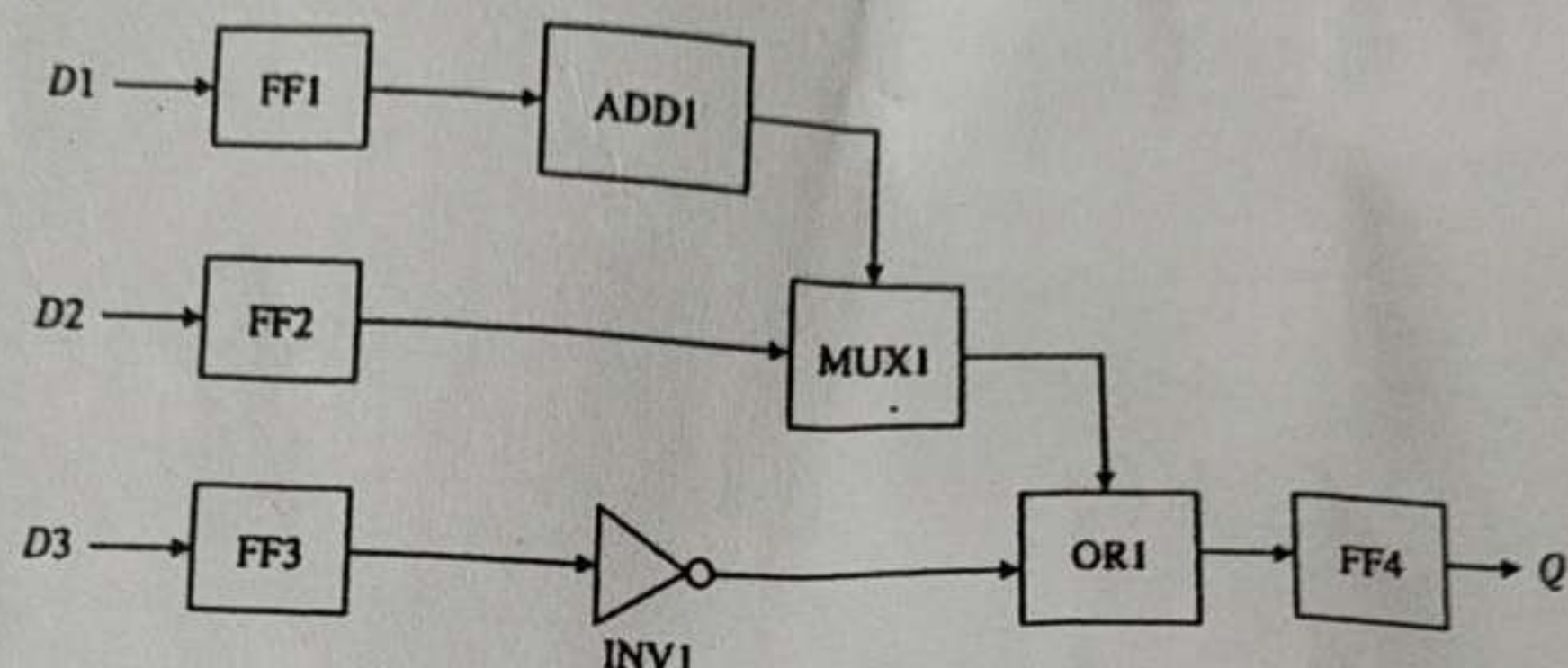


Fig.1

The maximum/minimum delays (t_{pd}/t_{cd}) are: ADD1 = 1.60/0.80 ns, MUX1 = 0.90/0.40 ns, INV1 = 0.30/0.15 ns, and OR1 = 0.70/0.30 ns. For all flip-flops, $T_{CQ,max}/T_{CQ,min} = 0.50/0.20$ ns and $T_{setup}/T_{hold} = 0.40/0.20$ ns. The clock period is 4 ns, launch/capture clock latencies are 0.30/0.70 ns, and setup/hold uncertainties are 0.20/0.10 ns. Assume all MUX paths are valid.

- Determine all FF-to-FF paths and identify the critical setup and hold paths. [3]
- Calculate the arrival time, required time, and slack for the critical setup path. State whether setup is met. [3]
- Calculate the arrival time, required time, and slack for the critical hold path. State whether hold is met. [3]
- Determine the minimum clock period and maximum operating frequency. [3]
- Write the required SDC constraints for clock period, setup/hold uncertainty, and launch/capture clock latency. [3]

Consider the following synchronous datapath in Fig.2:

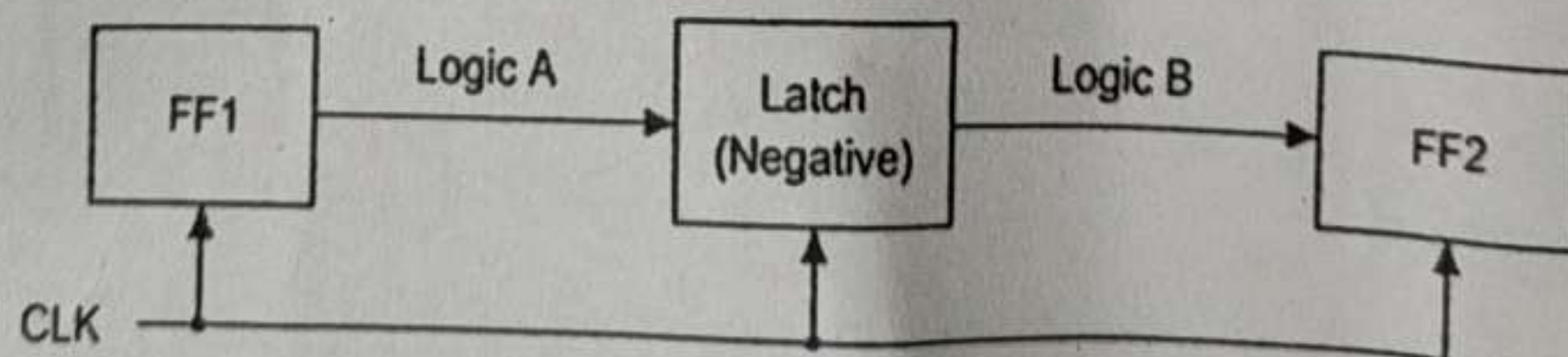


Fig.2

- 5 FF1, the latch, and FF2 operate from a common clock having a period of 10 ns and 50% duty cycle. The negative-level latch is transparent during the appropriate low phase of the clock. The following timing parameters are given:

$$\begin{aligned}
 T_{CQ,max} &= 0.5 \text{ ns}, T_{CQ,min} = 0.2 \text{ ns} \\
 T_{DQ,max} &= 0.2 \text{ ns}, T_{DQ,min} = 0.1 \text{ ns} \\
 T_{setup,L} &= 0.5 \text{ ns}, T_{hold,L} = 0.3 \text{ ns} \\
 T_{setup,FF2} &= 0.5 \text{ ns}, T_{hold,FF2} = 0.3 \text{ ns} \\
 T_{A,max} &= 6.5 \text{ ns}, T_{A,min} = 0.5 \text{ ns} \\
 T_{B,max} &= 8.0 \text{ ns}, T_{B,min} = 0.5 \text{ ns}
 \end{aligned}$$

Clock uncertainty is

and the OCV derates are

$$U_{\text{setup}} = 0.3 \text{ ns}, U_{\text{hold}} = 0.2 \text{ ns}$$

$$D_{\text{late}} = 1.10, D_{\text{early}} = 0.90$$

Assume zero clock latency.

- Determine the opening and closing instants of the negative-level latch and calculate the required OCV-adjusted early and late path delays. [3 Marks]
- Perform the setup and hold timing checks from FF1 to the latch and calculate the corresponding timing slacks. [4 Marks]
- Determine how much of the latch's transparent interval is utilized by the arriving data, and calculate the maximum permissible transparency usage considering setup time and uncertainty. [4 Marks]
- Continue the timing analysis through Logic B and perform the latch-to-FF2 setup check. Determine the data arrival time, required arrival time and setup slack at FF2. [3 Marks]
- Explain why the late OCV derate is applied for setup analysis whereas the early OCV derate is applied for hold analysis. [2 Mark]

*****All the best *****

Facult